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2 Real Analysis 2023

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Deferred Test 2

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Instructions:

- This question paper consists of **9 pages** (including this one).
- You have 90 minutes to answer the questions.
- Answer all questions in the corresponding boxes.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 50 points = 100% (52 points available).

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Problem 1

Let $(x_n)_{n=1}^{\infty}$ be the sequence defined recursively by

$$x_1 = 6,$$
$$x_{n+1} = \sqrt{3 + x_n}, \quad n \in \mathbb{N}.$$

(a) Prove that $2 \le x_{n+1} \le x_n \le 6$, $n \in \mathbb{N}$. (5 points)

Solution. We use induction in n. Notice that $x_2 = \sqrt{3+6} = 3$.

Induction base: We consider $n = 1$. We have $2 \le \underbrace{3}_{=x_2} \le \underbrace{6}_{=x_1} \le 6$.

Induction step: Assume that $2 \le x_{m+1} \le x_m \le 6$ for some $m \in \mathbb{N}$.
We have to show that $2 \le x_{m+2} \le x_{m+1} \le 6$.

We start with $2 \le x_{m+1} \le x_m \le 6$.

Add 3: $5 \le 3 + x_{m+1} \le 3 + x_m \le 9$.

Take the square root: $\sqrt{5} \le \underbrace{\sqrt{3 + x_{m+1}}}_{=x_{m+2}} \le \underbrace{\sqrt{3 + x_m}}_{=x_{m+1}} \le 3$.

We obtain $2 < \sqrt{5} \le x_{m+1} \le x_{m+2} \le 3 < 6$.
The induction step follows.

By the Principle of Mathematical Induction we conclude that $2 \le x_{n+1} \le x_n \le 6$ for all $n \in \mathbb{N}$.

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(b) Formulate the Monotone Convergence Theorem. (2 points)

Solution. If $(x_n)_{n=1}^{\infty}$ is a sequence that is increasing and bounded above, then it converges (to its supremum). If $(x_n)_{n=1}^{\infty}$ is a sequence that is decreasing and bounded below, then it converges (to its infimum).

(c) Explain how the Monotone Convergence Theorem is applied to show that the sequence $(x_n)_{n=1}^{\infty}$ converges. (2 points)

Solution. We have shown in (a) that $x_n \geq x_{n+1}$ for all $n \in \mathbb{N}$; this means that $(x_n)_{n=1}^{\infty}$ is decreasing. Moreover, we have shown that $x_n \geq 2$ for all $n \in \mathbb{N}$; this means that $(x_n)_{n=1}^{\infty}$ is bounded below. By the Monotone Convergence Theorem we conclude that $(x_n)_{n=1}^{\infty}$ converges.

(d) Find $\lim_{n\to\infty} x_n$. (3 points)

Solution. Let $x = \lim_{n\to\infty} x_n$.
Taking the limit as $n \to \infty$ in the recurrence $x_{n+1} = \sqrt{3 + x_n}$, we obtain that

$$x = \sqrt{3 + x}.$$

This equation is equivalent to the quadratic equation

$$x^2 - x - 3 = 0.$$

The solutions are $x = \frac{1}{2} \pm \sqrt{\frac{1}{4} + 3} = \frac{1 \pm \sqrt{13}}{2}$. However, $\frac{1-\sqrt{13}}{2} < 0$, and therefore it cannot be the limit.
We conclude that $\lim_{n\to\infty} x_n = \frac{1+\sqrt{13}}{2}$.

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Problem 2

(a) Consider the series $\sum_{n=0}^{\infty} \frac{1+2^n}{3^n}$. Find an expression for its partial sums. Use this expression to show that the series converges, and find its sum. (3 points)

Solution. For $n \in \mathbb{N}$ we have

$$s_n = \sum_{k=0}^{n} \frac{1+2^k}{3^k} = \sum_{k=0}^{n} \left(\frac{1}{3}\right)^k + \sum_{k=0}^{n} \left(\frac{2}{3}\right)^k = \frac{1-\left(\frac{1}{3}\right)^{n+1}}{1-\frac{1}{3}} + \frac{1-\left(\frac{2}{3}\right)^{n+1}}{1-\frac{2}{3}}$$
$$= \frac{3}{2}\left(1-\left(\frac{1}{3}\right)^{n+1}\right) + 3\left(1-\left(\frac{2}{3}\right)^{n+1}\right).$$

Since $\left|\frac{1}{3}\right| < 1$ and $\left|\frac{2}{3}\right| < 1$, we have that $\lim_{n\to\infty} \left(\frac{1}{3}\right)^{n+1} = 0$, $\lim_{n\to\infty} \left(\frac{2}{3}\right)^{n+1} = 0$. Thus,

$$\lim_{n\to\infty} s_n = \frac{3}{2} + 3 = \frac{9}{2}$$

exists. It follows that the series converges and

$$\sum_{n=0}^{\infty} \frac{1+2^n}{3^n} = \frac{9}{2}.$$

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(b) For each of the following series, determine whether it is convergent or divergent. Give a justification.

(i) $\displaystyle\sum_{n=1}^{\infty} \frac{(2n)^n}{n!}$ (2 points)

Solution. We apply the Ratio Test:

$$L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{2^{n+1}(n+1)^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n n^n} = \lim_{n\to\infty} \frac{2(n+1)(n+1)^n}{(n+1)n^n}$$
$$= 2\lim_{n\to\infty} \left(\frac{n+1}{n}\right)^m = 2\lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n = 2e > 1.$$

By the Ratio Test, the series diverges.

(ii) $\displaystyle\sum_{n=1}^{\infty} \left(\frac{n^2+2n-2}{3n^2+5n}\right)^n$ (2 points)

Solution. We apply the Root Test:

$$L = \lim_{n\to\infty} |a_n|^{\frac{1}{n}} = \lim_{n\to\infty} \frac{n^2+2n-2}{3n^2+5n} = \lim_{n\to\infty} \frac{1+\frac{2}{n}-\frac{2}{n^2}}{3+\frac{5}{n}} = \frac{1}{3} < 1.$$

By the Root Test, the series converges.

(iii) $\displaystyle\sum_{n=1}^{\infty} \frac{(\cos n)^2}{n+n\sqrt{n}}$ (2 points)

Solution. We use the Comparison Test. For any $n \in \mathbb{N}$ we have

$$0 < \frac{(\cos n)^2}{n+n\sqrt{n}} \le \frac{1}{n+n\sqrt{n}} \le \frac{1}{n\sqrt{n}} = \frac{1}{n^{\frac{3}{2}}}.$$

The series $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$ is a convergent p-series with $p = \frac{3}{2}$. By the Comparison Test, the series $\sum_{n=1}^{\infty} \frac{(\cos n)^2}{n+n\sqrt{n}}$ converges.

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(c) For each of the following series, determine whether it is absolutely convergent, conditionally convergent or divergent. Give a justification.

(i) $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n}+1}$ (4 points)

Solution.
First we consider the series $\sum_{n=0}^{\infty} \frac{1}{\sqrt{n}+1}$. For any $n \in \mathbb{N}$ we have

$$\frac{1}{\sqrt{n}+1} \geq \frac{1}{\sqrt{n}+\sqrt{n}} = \frac{1}{2\sqrt{n}}.$$

The series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ is a divergent p-series, hence also the series $\sum_{n=1}^{\infty} \frac{1}{2\sqrt{n}}$ diverges. By the Comparison Test, the series $\sum_{n=0}^{\infty} \frac{1}{\sqrt{n}+1}$ diverges.
For the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n}+1}$ it means that it does not converge absolutely.
The series $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n}+1}$ is an alternating series. We have $\lim_{n\to\infty} \frac{1}{\sqrt{n}+1} = 0$, and $\frac{1}{\sqrt{n}+1} \geq \frac{1}{\sqrt{n+1}+1}$ for all $n \in \mathbb{N} \cup \{0\}$. By the Alternating Series Test, the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n}+1}$ converges.
We conclude that the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n}+1}$ converges conditionally.

(ii) $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ (3 points)

Solution.
We consider the series $\sum_{n=0}^{\infty} \frac{1}{n!}$. We have

$$L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{n!}{(n+1)!} = \lim_{n\to\infty} \frac{1}{n} = 0 < 1.$$

Hence, the series $\sum_{n=0}^{\infty} \frac{1}{n!}$ converges by the Ratio Test.
We conclude that the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ converges absolutely.

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Problem 3
Read the following theorem and an outline of its proof.

Theorem. Let $K \subseteq \mathbb{R}$. If K is closed and bounded, then K is compact.

Proof. Let$(x_n)_{n=1}$ be a sequence in K. Since $(x_n)_{n=1}^{\infty}$ is bounded, it has a convergent subsequence. Since K is closed, the limit of this subsequence belongs to K, and the statement follows.

Answer the following questions:

(a) Give a definition of closed set. (1 points)

Solution. A set $K \subseteq \mathbb{R}$ is called closed if it contains all its limit points.

(b) Give a definition of bounded set. (2 points)

Solution. A set $K \subseteq \mathbb{R}$ is called bounded if there exist numbers $m, M \in \mathbb{R}$ such that $m \leq x \leq M$ for any $x \in K$.
An alternative definition: A set $K \subseteq \mathbb{R}$ is called bounded if there exists a numbers $M > 0$ such that $|x| \leq M$ for any $x \in K$.

(c) Give a definition of compact set. (1 points)

Solution. A set $K \subseteq \mathbb{R}$ is called compact, if every sequence in K has a subsequence that converges to an element of K.

(d) How do we conclude that $(x_n)_{n=1}^{\infty}$ has a convergent subsequence? Name and formulate the corresponding theorem. (1 points)

Solution. This follows by the Bolzano-Weierstrass Theorem: Any bounded sequence has a convergent subsequence.

(e) How do we conclude that the statement follows at the end of the proof? (2 points)

Solution. We took an arbitrary sequence $(x_n)_{n=1}^{\infty}$ in K and showed that it has a subsequence that converges to a point in K. This is exactly the definition of compact set. We conclude that K is compact.

(f) Give an example of a set which is compact. Justify that this set is compact. (2 points)

Solution. For example, the set $[0, 1]$ is compact. It is a closed interval, and we know that a closed interval is a closed set. It is bounded, since $0 \leq x \leq 1$ for all $x \in [0, 1]$. By the theorem above, $[0, 1]$ is closed.

(g) Give an example of a set which is not compact. Justify that this set is not compact. (2 points)

Solution. For example, the set $(0, 1)$ is not compact. This follows from the fact that it is not closed: it does not contain its limit point 0.

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Problem 4
Consider the set $S = \left\{\frac{(-1)^n n}{n+1} : n \in \mathbb{N}\right\} \cup \{1\} \cup [2, 3]$.

(a) Determine, without proof, the set S' of all limit points of S. (1 points)

Solution. $S' = \{-1\} \cup \{1\} \cup [2, 3]$.

(b) Determine, without proof, the closure $\overline{S}$ of S. (1 points)

Solution. $\overline{S} = \left\{\frac{(-1)^n n}{n+1} : n \in \mathbb{N}\right\} \cup \{-1\} \cup \{1\} \cup [2, 3]$.

(c) Determine, without proof, the set of all isolated points of S. (1 points)

Solution. The set of isolated points of S is $\left\{\frac{(-1)^n n}{n+1} : n \in \mathbb{N}\right\}$.

(d) Is S compact? Justify your answer. (2 points)

Solution. A subset of $\mathbb{R}$ is compact if and only if it is closed and bounded.
The set S is not closed: it does not contain its limit point -1.
We conclude that S is not compact.

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Problem 5

(a) Let $f : A \to \mathbb{R}$ be a function, let a be a limit point of A and $L \in \mathbb{R}$. Define what it means that $\lim\limits_{x\to a} f(x) = L$. (2 points)

Solution. By definition we say that $\lim\limits_{x\to a} f(x) = L$ if

$$\forall\, \varepsilon > 0\ \exists\, \delta > 0\ \forall x \in A : 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

In other word, for any $\varepsilon > 0$ there exists $\delta > 0$ such that $|f(x) - L| < \varepsilon$ for all $x \in A$ with $0 < |x - a| < \delta$.

(b) Let $f : \mathbb{R} \to \mathbb{R}$. Suppose that $\lim\limits_{x\to 2} f(x) = 3$. Prove by definition that

$$\lim_{x\to 2} \frac{f(x)+1}{f(x)-1} = 2.$$

(6 points)

Solution. We have

$$\left|\frac{f(x)+1}{f(x)-1} - 2\right| = \left|\frac{f(x)+1-2f(x)+2}{f(x)-1}\right| = \left|\frac{-f(x)+3}{f(x)-1}\right| = \frac{|f(x)-3|}{|f(x)-1|}.$$

By the definition of the limit $\lim_{x\to 2} f(x) = 3$ there exists $\delta_1 > 0$ such that

$$0 < |x-2| < \delta_1 \implies |f(x) - 3| < 1.$$

For such x we have $-1 < f(x) - 3 < 1$, and $2 < f(x) < 4$. We conclude that $1 < f(x) - 1 < 2$, and $|f(x) - 1| > 1$.

Thus, for $0 < |x - 2| < \delta_1$ we have

$$\left|\frac{f(x)+1}{f(x)-1} - 2\right| = \frac{|f(x)-3|}{|f(x)-1|} < |f(x) - 3|.$$

Given $\varepsilon > 0$, there is $\delta_\varepsilon > 0$ such that

$$0 < |x-2| < \delta_\varepsilon \implies |f(x) - 3| < \varepsilon.$$

Put $\delta = \min\{\delta_1, \delta_\varepsilon\} > 0$. We have

$$0 < |x-2| < \delta \implies \left|\frac{f(x)+1}{f(x)-1} - 2\right| < |f(x) - 3| < \varepsilon.$$

By definition, $\lim_{x\to 2} \frac{f(x)+1}{f(x)-1} = 2$.